

Colonial Medicine and Its Legacies

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The groundswell of interest in global health issues over the past few years sometimes causes observers to see the field as “new.” There are indeed many new features that characterize global health concerns in the twenty-first century, such as the worldwide visibility of the AIDS pandemic and the reshaping of global biosurveillance in the wake of the SARS and H1N1 epidemics. Many of the therapeutics in the medical arsenal are also new and far better evaluated than their predecessors. But transnational and pandemic diseases are not new, even when the ability to track them—for some, an example of biopower—becomes more sophisticated or involves new diagnostics for previously undescribed (or truly novel) pathogens.

The new, as always, is rooted in the old, and serious biosocial exploration of global health today would be incomplete without plumbing history. Even a cursory evaluation of current global disparities in disease burden and access to biomedical therapeutics makes clear that efforts to improve global health equity must navigate a landscape littered with the wreckage (and the occasional glorious monument) of programs past. In turn, many of these failures—such as the global malaria eradication campaign of the 1950s and 1960s (discussed in this chapter)—may be attributed, in part, to a lack of historical reflection and biosocial analysis. Like the social theories considered in the previous chapter, history can help us understand the intended and unintended consequences of global health interventions.¹

The term “global health” was coined to define health problems and interventions extending beyond national boundaries, including those between developed and developing countries.² As noted in chapter 1, the term is distinct from “international health,” used throughout the twentieth century to describe efforts to improve the health of populations transnationally—usually from global North to global South, and often grounded in development programs with diverse (and sometimes hidden) agendas. It is also distinct from “colonial medicine,” the nineteenth-century term that described medicine in the days of imperial rule and colonization.³ As this chapter explains, the legacy of colonial medicine has a long reach: both global empires and the institutions of colonial medicine persisted well into the second half of the twentieth century and, in some regards, persist today. Even the process of identifying and ranking health challenges—what historians of science call *problem choice*—demonstrates that global health priorities in the present have been patterned by social forces with roots in the colonial past.

In this chapter, we first briefly trace the relationship between global health and empire, exploring how colonial institutions exerted power over indigenous populations by adjudicating health status and care. Second, we describe how global commerce and international relations became enmeshed with global health and examine the specialized and technocratic institutions that were set up to manage public health and colonial medicine, institutions that were the predecessors of today’s global public health authorities. Finally, by investigating a few key global health efforts and drawing on our framework of social theory, we reveal continuities that persist from colonial medicine and analyze the limitations of humanitarian models, old and new, of global health.

GLOBAL HEALTH AND GLOBAL EMPIRE

Notions of global health have influenced imperial ambitions, international relations, and global commerce for millennia. A concept of global health surely motivated hygienic reformers of the Roman Empire—when it constituted, at least in its own view, most of the known world—to standardize aqueducts and sewer systems and seek management of pestilential diseases across its many provinces.⁴

Since the fall of Rome, there no doubt have been countless attempts across the world to improve transregional health—and especially to stop plagues that disrupted commerce. There is no true start date for imperial medicine. There were empires in Asia and in Europe (and

in the lands between them), just as there were in the continents that would later be called the Americas. But the more direct forebears of international, and thus global, health can be found among European colonialists.

It is no accident that the redefinition of public health and biomedicine as scientific professions coincided with the moment at which European powers began to build empires. Most narratives describe medical and public health advances originating in the metropolises of Europe and North America and diffusing later to the peripheries of global empires. In a few instances, however, the dynamics of knowledge transfer were more complex, as exemplified by the discovery of quinine to treat malaria. Many of the attributes of both modern medicine and public health grew out of the unintended consequences of the globalization of science, commerce, and politics in the mid- to late nineteenth century. The history of colonial medicine shows that the sites of imperial occupation often served as laboratories for medical strategies later taken up by the colonizers.

Health was a central concern for European imperial projects from the first seaborne expeditions to the New World, Africa, and Asia.⁵ This was in part a result of the devastating mortality associated with the Columbian Exchange—the trade, intended and unintended, of plants, animals, and diseases between the Eastern and Western hemispheres following the voyage of Christopher Columbus. Europeans were exposed to novel pathogens and, in turn, brought many with them to the newly explored lands. Differential vulnerability to epidemic disease informed the logistics of imperial expansion, and racial ideologies were used to justify empire.

On a material level, differences in disease susceptibility between colonizer and colonized alternately aided and threatened plans for imperial expansion. Historian Alfred Crosby coined the term “ecological imperialism” to describe the exchange of organisms triggered by exploration and conquest.⁶ In Haiti, for example, several hundred thousand indigenous Taíno Indians lived on the island of Hispaniola before the Spaniards arrived in 1492; by the end of the seventeenth century, when the island was divided between France and Spain, not a single native had survived. European conquest and “virgin-soil epidemics” of measles, smallpox, and tuberculosis, which spread usually by chance and sometimes by design, killed millions of American natives, from the Caribbean and across the continents and the isthmus that linked them. These sharply differentiated mortality rates were not seen as apoca-

lyptic by the colonizers: as late as 1763, British officials were handing out blankets purposefully infected with smallpox among American Indians. European settlers had immunity, or partial immunity, to some of these diseases and therefore were able to propagate them among indigenous populations they planned to subjugate.⁷

History is rich with examples of colonial projects that undermined the health of indigenous populations. The sleeping sickness epidemics that struck East Africa at the turn of the twentieth century were linked to dramatic shifts in the movement of people and livestock that occurred under colonial management.⁸ Connected along lines of intra-imperial transit, plague epidemics found new routes, such as the Indian Ocean shipping lanes of the British Empire in the first decades of the twentieth century.⁹ The British colonial and military propagation of the opium trade between India and China had deleterious effects on individuals and populations—another example of the unintended consequences (addiction, an increase in piracy, overt conflict) of purposive social action (bringing vast stretches of land under British imperial sway).¹⁰

Early colonists from New England to Patagonia came to interpret the disparity in infectious-disease mortality as a providential sign of the rightness of the European imperial project and evidence of the frailty of “savage” bodies compared to European ones.¹¹ Over time, this observed disparity hardened into racial hierarchies based on embodied and seemingly unalterable biological characteristics.¹²

Colonial Medicine

And yet the European body was not universally hardy. In the second half of the nineteenth century, European cartographers elided “uncharted” territories to depict the world as the shared property of a handful of empires. In the early 1800s, European influence in many areas of Africa and Asia had been limited to fortified coastal settlements and trading zones, especially in tropical zones such as the “white man’s grave” of the Gold Coast of West Africa. This sorry moniker referred to the staggering European mortality rates of 300 to 700 deaths per 1,000 population in the first year of settlement.¹³ Multiple expeditions attempting to penetrate Africa’s interior had failed, their members decimated by disease. As late as 1841, a missionary expedition of 150 Africans and 150 Europeans was launched up the Niger River to “civilize” the people of the interior. Forty-two of the Europeans succumbed to malaria;

none of the Africans did. In 1854, Dr. William Baikie finally led the first successful European voyage up the Niger by using quinine to treat tropical fevers.¹⁴

While the role of quinine in enabling the military occupation of the tropics has perhaps been overstated, the lingering narrative of the “white man’s grave” is instructive on at least two levels.¹⁵ First, it illustrates the link between colonial medicine and imperial conquest. Second, it illustrates the use of “the tropics” as a laboratory and a source of test subjects for medical and public health research and practice.¹⁶

Many scholars have noted the ways in which colonial medicine facilitated the expansion of European settlements in West Africa.¹⁷ Quinine, still used today to treat certain forms of severe malaria, aided European exploits in the tropics. Perhaps one of the lesser-known products of the age of exploration was the introduction of the bark of the cinchona tree—originally known only within the area of modern-day Peru—into the pharmacopeia of European physicians. Cinchona bark had been used for centuries by South American peoples to treat fevers, but the trees were scarce. The fever-fighting bark became the object of fierce skirmishes among European powers who saw access to the cinchona tree as key to their military success within pestilential tropical zones.¹⁸ In 1820, French chemists Pierre-Joseph Pelletier and Joseph Bienaimé Caventou isolated quinine from cinchona bark and demonstrated that it was the active ingredient against tropical fevers. The isolation of quinine from cinchona bark was an early example of biomedical therapeutics: isolating an active ingredient from a botanic source.¹⁹

Colonial medicine was often cited as one of the virtues of the imperial enterprise, even long after other defenses of colonialism had been discarded. One champion of colonial medicine was Hubert Lyautey, a key strategist in the French invasions of the lands they called Indochine and Madagascar and the first resident-general of the new French possession of Morocco. He famously proclaimed that “the only excuse for colonization is medicine.”²⁰ Acknowledging the brutality of the colonial project, he nonetheless insisted that if there was one thing that “ennobles it and justifies it, it is the action of the doctor.”²¹ Reflecting on his long career as a colonial administrator, Lyautey noted in 1933 that “the physician, if he understands his role, is the most effective of our agents of penetration and pacification.”²²

And yet the project of colonial medicine did not merely serve the Promethean function of handing down the miracle of modern Western

medicine to colonial subjects. Colonial medicine originated to support the military, before broadening to include European-born administrators and civilians, with services therefore concentrated in important ports and urban centers. Beyond that, colonial medicine expanded to protect the health of the laboring populations insofar as local labor was required to run the vast plantations and mines that extracted economic resources for colonial interests.

Britain's Colonial Medical Service, for example, was charged with building and staffing clinics in particular areas of the empire, recruiting physicians trained at home to work in the colonies. As the service slowly expanded from urban centers, where larger hospitals were likely to be located, central administrators coordinated the activities of medical officers in the outlying districts. With varying degrees of proficiency, the facilities offered curative medicine, organized public health campaigns, and collected data regarding epidemics and other health indicators (see figure 3.1). Accounts from medical officers describe extensive paperwork and recordkeeping as well as the need for "private practice" outside official duties in order to supplement their incomes.²³ Even as the mission "to develop and protect" expanded later in the colonial era, colonial medical services tended to focus heavily on particular epidemic diseases.²⁴

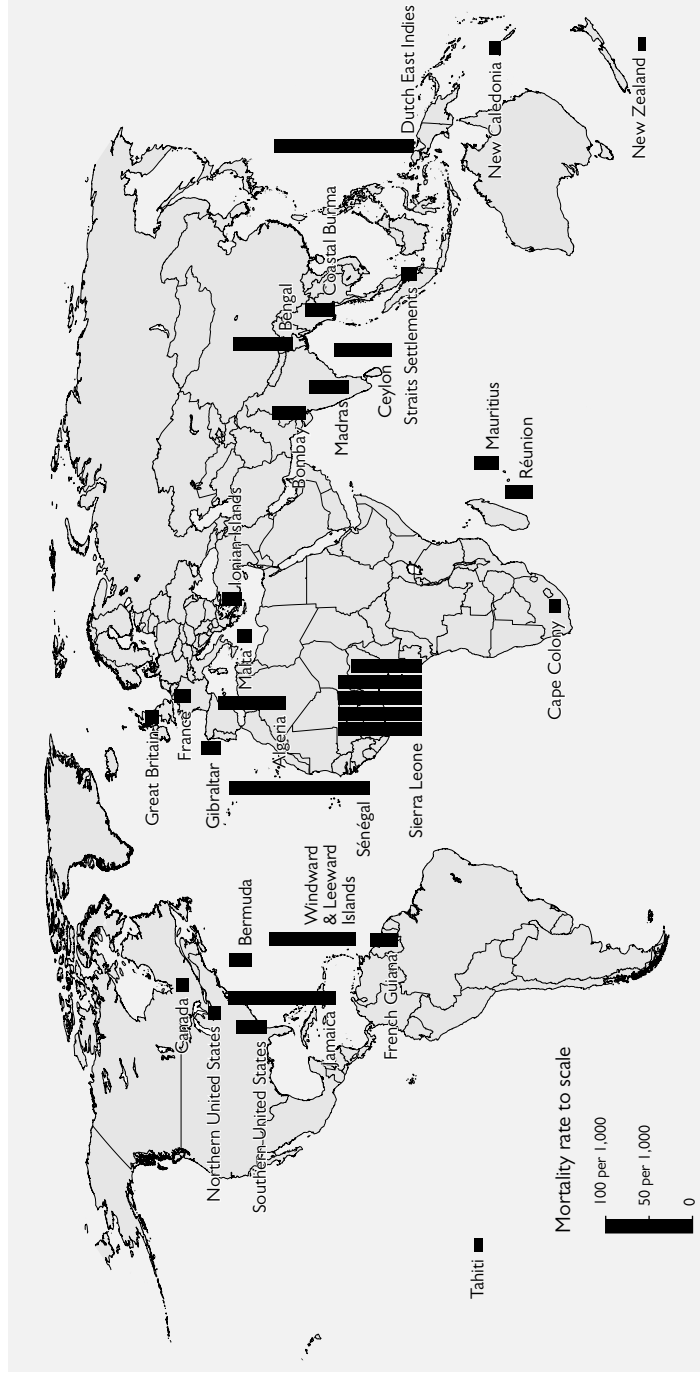
Indeed, links between global health and global security were established early on. Colonial medicine in its military guise prompted some of the first sustained international epidemiologic investigations, which compared bodies in health and disease across all continents and led to widespread use of the term "tropical medicine" (discussed in the following section). For example, in 1835, the British Empire commissioned a statistical study of mortality rates among troops of European descent stationed throughout the world, which found that death rates ranged from 11.5 per 1,000 for soldiers stationed in the United Kingdom to nearly 500 per 1,000 for those stationed in West Africa (see map 3.1). African-born troops deployed within the same latitude—from West Africa to Jamaica, for example—did not experience significant alterations in mortality. These statistics, once tabulated and circulated broadly, helped give credence to the belief that the black body was better suited for labor in hot climates than the white body. This belief contributed to the justification of the trans-Atlantic slave trade, which had enslaved roughly 11.4 million Africans by 1870.²⁵

In addition to being perceived as hardier under tropical conditions, darker bodies were also described by colonial administrators as vec-



FIGURE 3.1. A colonial medical officer on Buruma Island takes a sample of blood from an individual suffering from sleeping sickness (from the 1902 Uganda Sleeping Sickness Commission). Courtesy Wellcome Library, London.

tors of disease.²⁶ In the late nineteenth and early twentieth centuries, the field of “imperial hygiene” focused increasingly on the “uncivilized” and “unclean” practices of nonwhite subjects, whose “primitive” state made them a menace to the civilized world. Perceptions of cholera demonstrate this rhetoric of difference as blame: its fierce waves of epidemics over the course of the nineteenth century were seen to erupt periodically from the populous centers of India’s eastern seaboard and spread across Central Asia and into Europe. As British observer W. W. Hunter noted in 1872, the “over-crowded, pest-haunted dens around



MAP 3.1. Mortality rates among troops of European descent throughout areas of colonial expansion, 1817–1838. Before medical advances made treatment of common tropical diseases possible, European troops faced major losses because of deaths from infectious diseases. Source: Philip D. Curtin, *Death by Migration: Europe's Encounter with the Tropical World in the Nineteenth Century* (Cambridge: Cambridge University Press, 1989), 12.

Jagannath” in the eastern Indian city of Orissa were “at any moment, the centre from which the disease radiates to the great manufacturing towns of France and England.” Though the Indian pilgrims might “care little for life or death . . . such carelessness imperils lives far more valuable than their own.” Hunter continued:

One of man’s most deadly enemies has his lair on this remote corner of Orissa, ever ready to rush out upon the world, to devastate households, to sack cities, and to mark its line of march by a broad black track across three continents. The squalid pilgrim army of Jagannath with its rags and hair and skin freighted with vermin and impregnated with infection, may any year slay thousands of the most talented and beautiful of our age in Vienna, London, or Washington.²⁷

This concern with the links between distant, wealthy lands and diseased ones continues to shape conversations about global biosecurity today, including international dialogues about avian flu or SARS.²⁸ The recent, explosive epidemic of cholera in Haiti traces an ancient trail—from cholera-endemic regions of South Asia to the region whose natives were wiped out shortly after the arrival of Christopher Columbus and whose current inhabitants, now being felled by cholera, are the descendants of the very people held to be so resistant to tropical maladies.

The Birth of Tropical Medicine

The professionalization of colonial medicine kept pace with the development of vast empires. By the end of the century, few places remained beyond the reach of the ecological, economic, or territorial imperialism of the European, Russian, and Japanese empires.²⁹ No episode reflects this process as clearly as the 1884 Berlin Conference, in which European powers divided ownership of the entire “Dark Continent” among themselves with the stroke of a pen.

In addition to feeding notions of “the tropics” in the Western cultural imagination and codifying notions of racial difference, colonial health practices contributed to the construction of tropical medicine as a distinct discipline of medical research and practice. In the late nineteenth century, germ theory, elaborated by Louis Pasteur and Robert Koch, transformed notions of disease etiology by relocating the causes of illness from various “ill humors” (among other things) to microscopic agents of disease that came to infect the afflicted.³⁰ As cholera epidemics jeopardized European trade in North Africa, Koch, who had recently discovered the bacillus that causes tuberculosis, joined a series

of expeditions to Egypt and India to identify the cholera pathogen. Through study of the intestinal tissue of cholera patients and autopsies of cholera victims, Koch ultimately succeeded in isolating *Vibrio cholerae* in 1883.³¹

By the turn of the century, Patrick Manson and other clinicians had firmly decoupled “tropical medicine” and “cosmopolitan medicine”; the latter concerned diseases like tuberculosis that could be found anywhere in the world.³² Tropical diseases are largely associated with specific latitudes and regions; many are transmitted by insect vectors and caused by parasitic agents. Manson’s principles were swiftly taken up by colleagues such as Ronald Ross, a British-trained surgeon working in the Indian Medical Service, whose description of the role of the *Anopheles* mosquito in the life cycle of the malaria parasite won him a Nobel Prize in 1902. The field of tropical medicine flourished over the course of the twentieth century and led to the identification of scores of pathogens and vectors responsible for the scourges that afflicted poor people living in hot climates. But as many have noted, temperature, humidity, and latitude are rarely, if ever, the sole determinants of the distribution of classic tropical maladies.³³ The history of tropical medicine helps explain why in rich countries the phrase “global health” connotes diseases of “elsewhere”—problems affecting an othered “them” rather than an inclusive “us.”

The success of the new discipline of tropical medicine related directly to the shifting logics of imperial governance. Near the close of the nineteenth century, Joseph Chamberlain, secretary of state for the British colonies, advanced a “constructive imperialism”—the “exploitation” of vast “underdeveloped estates,” which would require attending to the health needs of native and non-native laborers and British settlers. The new science of tropical medicine suggested that one could control the damaging economic effects of epidemic disease by fighting its nonhuman vectors (such as the *Anopheles* mosquito) without providing direct curative services to native populations. This logic resonated within institutions of colonial medicine that tended to deal with native subjects as populations rather than as individuals. Manson was named medical advisor to the Colonial Office and established schools for tropical medicine in London and Liverpool.³⁴ At the graduation of the first class from the London School of Tropical Medicine, Manson summed up the valuable contribution of tropical medicine to the continuing European presence in the tropics, noting, “I now firmly believe in the possibility of tropical colonization by the white races. . . . Heat

and moisture are not in themselves the direct cause of any important tropical disease. The direct causes of 99 percent of these diseases are germs. . . . To kill them is simply a matter of knowledge and the application of this knowledge.”³⁵

While in theory a new paradigm of etiology—shifting the locus of disease from the “diseased native” to the microorganism—might have deracialized the discourse surrounding infectious disease, in practice the opposite was often true (as Hunter’s description of Indian cholera epidemics demonstrates). Germ theory did introduce a nonhuman target for disease-control efforts, but it also introduced a new vector—the “healthy carrier”—whose hygienic practices were as important to disease control as traditional measures. The prototypical “healthy carrier” was Mary Mallon, “Typhoid Mary,” a New York-based “Irish” cook known to have infected at least fifty-three people with typhoid fever in the first two decades of the twentieth century. Although she showed no symptoms of her *Salmonella typhi* infection, Mallon was branded a threat to society and spent the last twenty-three years of her life incarcerated by the State of New York.³⁶ Worse fates befell colonial subjects blamed (rightly or wrongly) for causing disease among white populations.

Warwick Anderson’s historical research reveals the repressive public health measures used during the U.S. military occupation of the Philippines from 1898 to 1912.³⁷ Americans had long associated the Filipino lifestyle with backward morals and unhygienic behavior and blamed Filipinos for devastating cholera epidemics, in which more American soldiers died than in the entire Spanish-American War. When a 1902 epidemic killed two hundred thousand people in the U.S.-occupied Philippines, the U.S. Army Public Health Force declared a “cholera war” that razed villages, administered drugs—effective and (more often) ineffective—by force, imposed quarantines, and seized and cremated afflicted bodies.³⁸ Draconian actions such as these contributed to turning contemporary intellectuals William James and Mark Twain against American imperialism.

In this context, Anderson argues, the asymptomatic carrier of cholera microbes “recast the constitutional dangers of tropical climates into a form that stressed the hazards of a parasitic environment, a biological and social terrain in which the salients were the Filipino bodies containing invisible microbes.”³⁹ When it was discovered that Filipinos were immune to some of the diseases that adversely affected foreigners, they were described as “microbial *insurrectos*”—an epidemiologic parallel to the armed insurgency—and as a direct threat to the health

of the American residents of the Philippines.⁴⁰ The science of tropical medicine, far from extinguishing a racialized language of the “diseased native,” enabled it. As one American medical officer in Manila noted, “as long as the Oriental was allowed to remain disease-ridden, he was a constant threat to the Occidental who clung to the idea that he could keep himself healthy in a small disease-ringed circle.” Instead, another colonial medical officer quipped that “blonds and brunettes now had more to fear from contact with a variety of diseased native fauna than from exposure to the rays of the tropical sun.”⁴¹

The management of cholera in the occupied Philippines provides an example of heavy-handed and racialized practices in global health—most of them completely ineffective, as far as the cholera bacterium was concerned—in the wake of germ theory. But even in less explicitly militarized encounters, a moral language of health, hygiene, and the “civilizing process” suffused colonial discourse (see, for example, figure 3.2) and was invoked to justify the continued imperial presence throughout the first half of the twentieth century. In stories, magazine articles, and advertisements, nonwhite colonial subjects were depicted as childlike or, worse, as part of the local flora and fauna that made the tropics a risky place for white bodies.⁴² Physicians, social scientists, and social theorists—quite apart from the overt eugenicists—were complicit. For example, the French anthropologist Lucien Lévy-Bruhl popularized a theory of “primitive mentality” that posited structural differences between the “primitive” and “Western” minds.⁴³ Years later, the Martinique-born Franco-African psychiatrist Frantz Fanon used Lévy-Bruhl’s metaphor as an example of how colonial medical practices perpetuate a sense of inferiority among the colonized.⁴⁴

Fanon’s writings helped to inspire a key strain of anticolonialism in twentieth-century intellectual and political history; in the wake of decolonization, his work formed the bedrock of a new canon of post-colonial theory. Yet the idea of a distinct “primitive mind” has persisted well into this century. In 2001, the head of the U.S. Agency for International Development (USAID), the lead U.S. funder of development efforts in what is now termed the developing world, decreed that antiretroviral therapy would fail in Africa because Africans “don’t know what Western time is.” He claimed that certain Africans would be unable to adhere to their medication courses, noting, “You say, take it at ten o’clock, they say, ‘What do you mean, ten o’clock?’”⁴⁵ It is worth noting that this U.S. official was merely putting into words, and publicly, ideas that were widespread among leading voices in interna-



FIGURE 3.2. Popular culture during the colonial era often portrayed exotic, racialized visions of the colonies and the “diseased natives,” as did this advertisement for Pears’ Soap, which exemplifies the moral language of health and the “civilizing process.” The advertisement appeared in *Harper’s Weekly* in September 1899.

tional health, who also argued, in less cosmological terms, that AIDS treatment was “too difficult” or “too complex” in the very continent most afflicted by the disease (as chapter 5 describes).

Missionary Medicine

European and North American perceptions of colonized peoples typically included moral characterizations. As diverse peoples confronted each other through trade and colonization, they were also exposed to new forms of religion. Though Christian missions to foreign lands are as old as Christianity itself, imperial expansion opened new possibilities for proselytizing. These efforts included medical missions, which in turn helped to shape the history of global health. Michael Worboys points out that while colonial states concentrated their health efforts in urban areas and focused on epidemic disease, missionaries engaged more directly with local populations.⁴⁶ For people in many parts of the

world, Christian medical missionaries were the sole point of contact with biomedicine. These interactions reveal that some of the roots of modern global health politics can be traced to the moral economies of missionary medicine.

Historian Megan Vaughan argues that, in addition to their different target populations, medical missionaries and colonial bureaucracies in Africa had different ideologies in approaching the “diseased native.” Aligned with the colonial discourse, many missionaries believed that indigenous religions and social systems were backward, immoral, and unclean. The missionaries, however, held up Western civilization and Christianity as a solution to illness and a pathway to salvation. In contrast, the medical institutions established by the colonial state (which we refer to as colonial medicine) at times tried to limit the extent to which African communities—especially in rural areas—were disrupted by Western culture and lifestyle.⁴⁷ For example, “detribalization,” or “acculturation,” was frequently cited as a cause of mental illness. Vaughan describes a 1935 report by two colonial medical officers in Nyasaland (now Malawi) who argued that “the central cause of insanity was ‘acculturation,’ brought about, in the main, by [Western] education.”⁴⁸

Whereas colonial medicine focused on populations, missionary medicine tended to focus on individuals. To illustrate this distinction, Vaughan challenges the application of Michel Foucault’s notion of biopower in the context of colonial medicine. Biopower (discussed in chapter 2) depends on the creation of active subjects who internalize and reproduce state definitions of their biologic selves. In African colonies, Vaughan argues, no real link was formed between population-level statistics and individual self-conceptions. Colonial states treated colonized peoples as collective Others—usually as tribes—rather than as productive individuals:

In contrast to the developments [in Europe] described by Foucault, in colonial Africa group classification was a far more important construction than individualization. Indeed, there was a powerful strand in the theories of colonial psychologists which denied the possibility that Africans might be self-aware individual subjects, so bound were they supposed to be by collective identities. If modern power operates through the creation of the “speaking subject,” then this colonial power cannot be the power which Foucault is describing.⁴⁹

Missionary medicine, however, was preoccupied with the reformation of individual souls and bodies. Clinical care at mission hospitals was an individualizing force, according to Vaughan, focused on per-

sonal illness, personal hygiene, and personal sin.⁵⁰ Missionary medicine, therefore, operated through mechanisms more akin to biopower than its colonial counterpart.

Within Africa, the expansion of medical missionary work in the late nineteenth and early twentieth centuries was overwhelmingly Christian-based and generally came from Protestant churches rather than the Catholic Church. Though Catholic missions had been established around the world, the fact that Protestant missionaries were not necessarily members of the clergy probably contributed to their greater numbers and greater rate of expansion. According to historian David Hardiman, Catholic missions' ability to expand health services was hampered because the church discouraged nuns from seeking medical training; in contrast, female nurses and physicians abounded in Protestant missions.⁵¹

Initially, medical missionaries were not always well-trained doctors; until the professionalization of the medical field in the mid-nineteenth century, missionaries whose formal training consisted only of a few lectures often engaged in clinical care. By the end of the century, however, medical missionaries were expected to be trained physicians familiar with theology as opposed to pious people with some medical experience. This professionalization coincided with advances in medical science, including the advent of germ theory and antiseptics, anesthesia, and early vaccines. Hence, mission doctors at the turn of the century were far more effective in their clinical work than their predecessors.⁵² The quality of these "deliverables"—in contrast to the imposed and ineffective attempts to stop cholera in the Philippines—began to rise even before the advent of antibiotics.

Just as disease was associated with backward morals of native peoples, traditional medicine was often linked to "heathen" religions. "Among all rude races," one missionary wrote, "magic and medicine are wedded, the priest and the doctor are one."⁵³ Clinical work was therefore often understood as more than treating illness; rather, many missionaries sought to convert patients both to Christianity and to their notions of modernity. On the one hand, missionaries hoped that the rational order of the clinic and the biomedical project would encourage patients to shake off traditional faiths.⁵⁴ On the other hand, extended hospital stays were seen as prime opportunities for conversion, and prayer and church services became central parts of the inpatient experience. In short, mission medicine "was not done for a purely medical purpose, but used as a beneficent means to spread Christianity."⁵⁵

FIGURE 3.3. “A Congo Child’s Appeal,” an advertisement that appeared in the *Medical Missionary Journal* in 1909, requesting donations from Britons for a missionary hospital. Courtesy Bodleian Library, University of Oxford, shelfmark Per 133 d.83.



Just as missionaries were frequently the first point of contact between Europeans and non-Western peoples, the news and writings the missionaries sent back home were commonly Europeans’ main source of information about the colonies.⁵⁶ Missions depended on fundraising in “home churches”; patrons who sponsored beds in the clinics learned about their occupants through missionaries’ letters home (see an example of a fundraising appeal in figure 3.3).⁵⁷ The medical missionaries themselves became iconic figures in Western understandings of the colonies. The heroic image of physician-explorer David Livingstone, for example, popularized the idea of a “civilizing mission” and the role of a clinician in that process.⁵⁸ David Hardiman writes of the many young physicians inspired by Livingstone’s fame and public writings to become medical missionaries.⁵⁹

The iconography of Western physicians in developing countries persists, as those familiar with the global health movement can attest. Megan Vaughan compares colonial and missionary hero-figures to modern images of European and North American doctors working in Africa.⁶⁰ This comparison raises difficult questions: What motivates an individual to exchange comfort and familiarity for a career ded-

icated to improving the well-being of distant populations? To what extent does the fact that some of these providers have far more effective tools—preventatives, diagnostics, therapeutics—than their forebears dreamed possible make such comparisons misleading? What is noble in these colonial legacies, and what is perilous, for global health practitioners and the populations they seek to help? Are the answers different when the stated purpose of the so-called modern missionaries is health equity and shoring up local health systems? How important is the distinction between building up missionary health facilities and strengthening local health authorities? Students and practitioners of global health encounter these questions every day.

GLOBAL HEALTH, GLOBAL COMMERCE, AND THE FOUNDATIONS OF INTERNATIONAL HEALTH BUREAUCRACIES

In reviewing the history of global health, we find that narratives are often closely tied to specific diseases. The history of sleeping sickness, for example, is limited to the “republic of the tsetse fly,” a wide belt across the waist of equatorial Africa.⁶¹ But the understanding of global health derived from studying such a geographically delimited disease can sometimes differ from the one that emerges in the history of more distributed plagues such as tuberculosis, malaria, or pandemic influenza. Here, we consider one such disseminated disease (cholera), as well as an endemic disease (yellow fever), to illustrate the intertwined histories of global health, global commerce, and biomedicine.

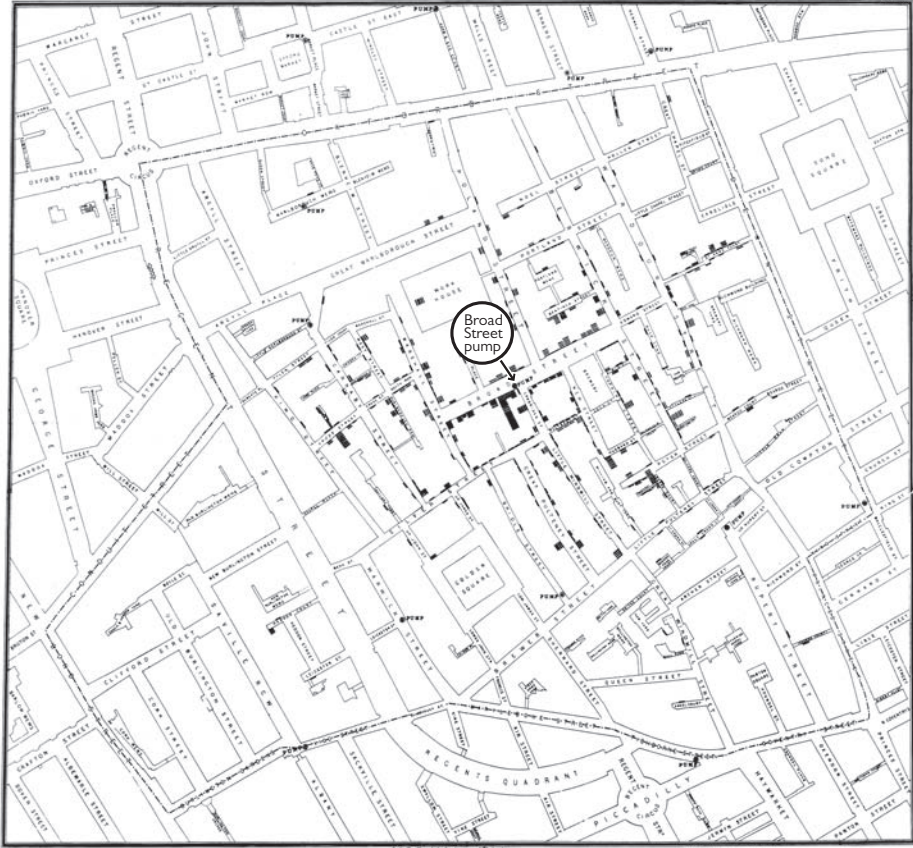
Perhaps no disease exemplifies these intertwined histories as clearly as cholera. A series of major cholera epidemics over the course of the nineteenth century was communicated directly and rapidly along new trade routes. The outbreaks thus constituted significant threats to the future of global commerce. The propagation of and response to these epidemics reflected the diffusion of new transportation and communication technologies, which now define the self-consciously global character of the early twenty-first century. New modes of communication such as the telegraph and the trans-Atlantic cable warned of local cholera epidemics by bringing news of outbreaks spreading elsewhere. As new feats of engineering, such as the building of the Suez Canal, decreased the time of transit between Madras and Marseilles, for example, global commerce became an increasingly agile potentiator of the pandemic reach of infectious diseases. The Office International d’Hygiène Publique (OIHP), one of the earliest permanent interna-

tional health bureaucracies, originated from a series of international sanitary conferences intended to address the vexing problem of cholera and international trade. As these conferences stretched across the late nineteenth and early twentieth centuries, almost all of them would ultimately focus on cholera.⁶²

Most of these conferences produced more debate than resolution, with little agreement on the etiology of cholera. The theory of “miasma”—that exposure to unsanitary environments, in particular to “bad air” filled with noxious mists or vapors, is responsible for the spread of disease—was the most broadly accepted explanation of contagious diseases during that era. In European cities, the theory of miasma came to justify many early hygienic approaches to public health, such as the civic engineering of tenements and sewer systems designed to remove offensive odors from susceptible populations. In tropical colonies, miasmatic theory came to define the geography of social stratification in colonial settlements: European-born populations settled in the more salubrious hilltop areas, where fresh breezes kept the environment healthy, while native-born populations were relegated to the fetid, swampy lowlands associated with fevers and disease.⁶³

After his now-famous investigation of a London cholera outbreak in 1854, anesthesiologist and amateur epidemiologist John Snow sought a meeting with the local health authority, the Board of Governors and Directors of the Poor. Snow presented the account of his investigation of the cholera epidemic in the neighborhood surrounding Broad Street, which suggested that the public pump was the epicenter of the epidemic and that the water it produced had likely been contaminated. Cases of cholera seemed to cluster around the pump (see Snow’s drawing, map 3.2), and Snow’s interviews with the families of those who were ill indicated that for most of these households, the pump provided the most accessible source of water (see figure 3.4). The meeting concluded with an order that the handle of the Broad Street pump be removed, which was carried out the very next day.⁶⁴ Though the original epidemic was already declining, the intervention may well have prevented a new outbreak. The removal of this pump handle is often considered the first successful policy recommendation supported by evidence-based infectious-disease epidemiology.⁶⁵

The debate was far from over, however; discrepant claims of causality would be made for decades. On March 5, 1855, Snow testified before Parliament, whose members were debating the elimination of what some called the “offensive trades,” defined as “trades that



MAP 3.2. John Snow's drawing mapped the 1854 cholera outbreak in London, demonstrating that cholera cases clustered around the Broad Street pump, a water source that had become contaminated. [Scale: 30 inches to 1 mile.] Source: John Snow, "On the Mode of Communication of Cholera, 1855." Courtesy Ralph R. Frerichs, Department of Epidemiology, University of California, Los Angeles.

released foul-smelling, noxious fumes," such as bone boiling and tallow melting.⁶⁶ This sanitary reform movement was driven by the argument that "poisonous vapors, whether miasmas rising from marshes or from decomposing organic matter near human dwellings, were the main cause of disease, including epidemic cholera, which had killed tens of thousands of people in England since 1831."⁶⁷ Snow, however, had little regard for the miasma theory and said as much: "I have paid a great deal of attention to epidemic diseases, more particularly to cholera, and in fact to the public health in general; and I have arrived at the



FIGURE 3.4. A contemporary image evoking the dangers of the Broad Street pump and its role in the 1854 cholera outbreak in London. Courtesy Centers for Disease Control and Prevention Image Library.

conclusion with regard to what are called offensive trades, that many of them do not assist in the propagation of epidemic diseases, and that in fact they are not injurious to the public health.”⁶⁸

Subsequent developments in the new field of medical microbiology would confirm Snow’s suspicions. However, despite Snow’s arguments, political interests heavily influenced international cholera regulations. The first forty years of the International Sanitary Conference (from 1851 to 1892) were stymied by debates that reflected national trade priorities rather than sound scientific evidence regarding disease transmission and prevention.

A few decades later, international deliberations over commerce and health in relation to the Panama Canal would serve as midwife to the birth of the first official international health organization, the Pan American Health Organization (PAHO), which remains an important player in the arena of global health to this day. Linking the Atlantic and Pacific oceans was a sixteenth-century dream that remained stalled until the completion of the Panama Railway in 1855. But a railroad was a modest endeavor compared to creating a passage for ships through the narrow isthmus separating two oceans. Driven by confidence and ambition, the French began construction of the Panama Canal. The

cost was steep: between 1881 and 1889—the period of French activity—more than twenty-one thousand workers died, many of them from yellow fever or malaria.⁶⁹ The project foundered; the dream was defeated by epidemic disease, at least temporarily.

The idea of completing the Panama Canal regained momentum at the turn of the twentieth century. One of the chief French officials involved in the early construction began to press the U.S. government to continue the project, hiring a well-known U.S. lawyer, William Nelson Cromwell, to lobby for the cause. At the time, the United States had been considering building a canal across Nicaragua. Cromwell's job was to convince members of Congress that Nicaragua would be a dangerous place to build such a canal—a task he sought to accomplish by designing a stamp with an image of a long-dormant Nicaraguan volcano coming to life and using the stamps on informational leaflets mailed to every member of the Senate.⁷⁰ In 1902, three days after the senators received the leaflets, the United States announced its plans to complete the canal in Panama.⁷¹

To succeed in this project, the Americans had to accomplish what the French had not: keeping the workforce healthy.⁷² Pestilential fevers were killing thousands of the laborers dispatched to build the canal, as well as their bosses, including engineers and government officials (see figure 3.5). Lingering belief in the relative immunity of black bodies to tropical fevers led to the widespread relocation of Afro-Caribbean laborers to the Canal Zone, where they were carefully segregated from white laborers and engineers but fell prey to the fevers just the same.⁷³

Analogous to the work of Patrick Manson and Ronald Ross in Great Britain, Giovanni Battista Grassi in Italy, and Charles Alphonse Laveran in France, the Canal Zone soon became a public theater for the deployment of modalities of tropical medicine in the Western Hemisphere. Carlos Finlay, a Cuban physician-scientist (pictured in figure 3.6)—building on ideas set out by Pasteur, Koch, and others—sought to discover the microorganisms and vectors causing the dreaded “yellow jack” (yellow fever) that had long plagued Havana. By 1891, Finlay's research pointed to the mosquito as the responsible vector. Walter Reed, a U.S. Army physician in Cuba, confirmed the theory through a set of elegant experiments conducted at Camp Lazear in Cuba. Volunteers exposed to dirty linens and bedclothes of yellow fever victims—but protected from mosquito exposure—remained healthy in comparison to volunteers living in hygienic, well-ventilated conditions who were exposed to mosquitoes. Armed with this new evidence, General

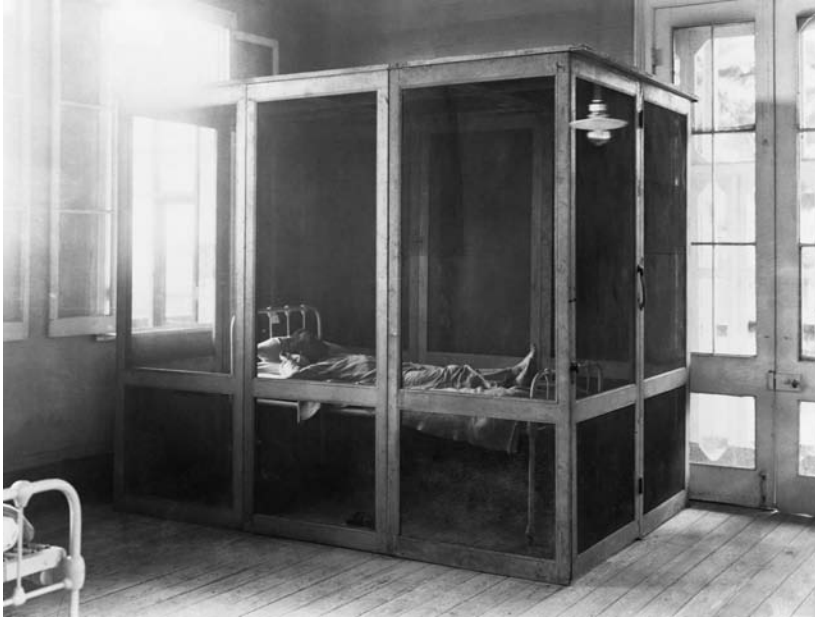


FIGURE 3.5. A yellow fever cage in Ancón Hospital, Panama, used to isolate victims of yellow fever during the construction of the Panama Canal. Courtesy CORBIS Images.

William Gorgas eliminated yellow fever from Havana through a program of mosquito control. Gorgas was then tapped to do the same in the Canal Zone.⁷⁴

Implementing the wide-reaching yellow fever elimination project required broad institutional and international support. Shortly after announcing the U.S. initiative to complete the Panama Canal, President Theodore Roosevelt convened the First General International Sanitary Convention of the American Republics, held in Washington, D.C., in December 1902; Finlay was one of four individuals appointed to the convention's organizing committee.⁷⁵ Discussions focused on the general impact of diseases on international trade, including quarantine, prevention, and shipping regulations, and led to the creation of the International Sanitary Bureau (ISB), later to become PAHO.⁷⁶

During the Second International Sanitary Convention, in 1905, H.L.E. Johnson, a U.S. committee member, explicitly acknowledged the strategic commercial interests driving the United States in the Panama Canal project and the subsequent Pan-American public health effort: "I feel sure that as a few months or years pass by, the diseases



FIGURE 3.6. Cuban physician and scientist Dr. Carlos Finlay, whose research identified the mosquito as the carrier of the microorganisms causing yellow fever. Courtesy U.S. National Library of Medicine.

which have stood in the way of the completion of the Panama Canal, which we might term the ideal of the President of the United States to accomplish, will be removed and that the great good to this country which is expected in health, wealth, and prosperity will flow from it.”⁷⁷ President Roosevelt himself continued to express “the greatest interest and confidence in the work of the sanitarians in the Isthmian Canal Zone.”⁷⁸ These predictions and aspirations were realized in 1905, as the ISB, led by U.S. Surgeon General Walter Wyman, indeed managed to eliminate yellow fever in Panama.⁷⁹ Work continued on the Panama Canal (see figure 3.7), which was completed by 1914, beating the target by two years.

Long after the Canal Zone workers had cleared out their lockers and been repatriated (often forcibly) to their countries of origin, the ISB continued to hold international health negotiations in the Western Hemisphere. Following a Monroe Doctrine model that recognized American spheres of influence, the institution was renamed the Pan American Sanitary Bureau (PASB) in 1923. Today it exists as the Pan American Health Organization.⁸⁰ PAHO has, over time, for-



FIGURE 3.7. Sanitary engineers from the health department laying tile drains during the construction of the Panama Canal. Courtesy U.S. National Library of Medicine.

tified working relationships, shared discursive tools and vocabulary, and standardized disease definitions and methods of surveillance in an effort to serve as “a model for transnational health promotion and information sharing.”⁸¹ Although not the first organization to attempt international cooperation around public health issues, PAHO was the first permanent institution of its kind and is now the world’s oldest international public health agency.⁸²

The creation of the ISB was, in many ways, a realization of Max Weber’s prediction that bureaucracies would come to be influential actors in modern society. In addition to PAHO, two other transnational organizations emerged in the early twentieth century: the Office International d’Hygiène Publique (OIHP), as noted earlier, and the League of Nations Health Committee. Both bodies attempted to coordinate international epidemiologic information and disseminate scientific advances. Neither organization achieved universal relevance, however, and both suffered from many of the same weaknesses as the League of Nations itself, including the overly optimistic prescriptions for global democracy that would become a hallmark of Wilsonian internationalism.



FIGURE 3.8. A Rockefeller Foundation microscopist gives local residents in Panama information about hookworm. Courtesy Rockefeller Foundation/National Geographic Stock.

While these projects did not officially involve the U.S. government, many of them received funding from the New York–based Rockefeller Foundation, the single largest funder of global health efforts in the first half of the twentieth century.⁸³ The establishment of the Rockefeller Foundation followed the success of the Rockefeller Sanitary Commission in eradicating hookworm in the United States, a campaign reaching back to 1909. While the Rockefeller Foundation’s focus in international health was long the eradication and prevention of infectious diseases (see figure 3.8), the early directors of Rockefeller international health programs sought to use biomedical interventions (such as anti-helminthic drugs for the treatment of hookworm) as levers to build sturdier and more cost-effective public health systems (such as the creation of hygienic waste-removal systems and clean water supplies). Thus the roots of much of the terminology and practice of global health a century later—for example, using “vertical,” or disease-focused, interventions to strengthen health systems—are far older than typically acknowledged.

Again we are brought back to Max Weber’s work. Over time, these public and private institutions normalized transregional public health

policymaking by diffusing the methodologies deemed capable of producing knowledge about health and disease. These technocratic bureaucracies created a sense of both rational-legal authority and stability that was new—a century ago—to the global health landscape. This stability allowed the organizations' assets to be passed down through time and also enabled greater specialization than had previously been possible. Facilitated by the rise of technical rationality, the emerging field of international health became more efficient and broadened in geographic scope and agenda.

At the same time, the nascent institutionalization of international health engendered the perception that centralized rules and processes were superior to local practices. Bureaucrats often found themselves trapped in Weber's famous "iron cage," finding it increasingly difficult as individuals to challenge the structures of institutional knowledge that had been shaped by social processes deemed scientific. While these institutions, from PAHO to Rockefeller, provided an opportunity to reach populations with a depth and breadth not previously possible, their power and effectiveness were sometimes limited by the values entrenched in the knowledge frameworks that colored their agenda and approach. Sometimes public health successes led to a certain "fetishization of process" that was neither nimble nor relevant. In Mexico, for example, the Rockefeller Foundation pursued a hookworm elimination campaign despite evidence of limited hookworm prevalence. The choice to embark on the hookworm program was made at upper levels of the Rockefeller Foundation bureaucracy, applying one-size-fits-all prevention and treatment techniques across Latin America.⁸⁴

At the same time, the Rockefeller Foundation's anti-hookworm campaign also provided examples of how bureaucratization can boost efficiency. During the campaign in the Philippines (1913–1915), the target population was divided into "units of intervention," with a specialized task force assigned to each unit to systematically treat all afflicted persons in the unit. To measure progress, the foundation employed standardized recording methods, which generated detailed records kept in a centralized database. These administrative tools enabled the foundation to track the progress and impact of the campaign and allowed the transfer of this administrative efficiency across regions, from Asia to Latin America.⁸⁵

Efficiency, however, came at the cost of comprehensiveness and, at times, the effectiveness that results from attention to local context in program design. Rockefeller Foundation records from the 1930s reveal

that officers purposely avoided addressing other diseases that could clearly be attributed to socioeconomic conditions. One such example was tuberculosis, judged to be beyond the scope of the organization because of its links with poverty and other large-scale social forces.⁸⁶ At the Pan American Sanitary Conference in 1946, hosted by PAHO, a delegate from Peru described a delousing campaign among indigenous groups that was successful because of the involvement of the local population and the deep consideration that had been given to the local context. The delegate's insight, however, would not become a central focus of public health for many years.⁸⁷

Many critical accounts have documented how the efforts of PAHO and the Rockefeller Foundation, whose programs were carried out in current and former European colonies, were vehicles for perpetuating knowledge frameworks that had taken shape within institutions of colonial medicine. Like other medical and development institutions of the early twentieth century, the Rockefeller Foundation sought to modernize "traditional," "backward," and "non-Western" cultures. It focused on problems amenable to technical interventions like vector control (for hookworm, yellow fever, and malaria, for example), pursuing a series of mass eradication campaigns against diseases. These endeavors yielded some remarkable successes. But they also relied on a narrow view of international health as a field concerned with populations rather than with individuals, devoting its energies to campaigns focused exclusively on diseases instead of achieving a comprehensive state of good health.⁸⁸ The apparent ease with which John Snow moved between his clinical practice (alleviating pain, largely) and his epidemiologic and public health efforts (protecting populations) became elusive. An emerging rigidity between clinical medicine and population-based health care efforts became readily discernible, and it would come to dominate—and ultimately undermine—international health.

Although the rise of formal global health bureaucracies in the early twentieth century was important for standardization, efficiency, and transregional cooperation, these developments left little room for consideration of community involvement or other locally specific social and economic factors. The language and concepts of global health equity were overshadowed by narrow and, in the emerging paradigm, "cost-effective" interventions that were designed to address the health problems of "backward" peoples who needed a dose of modernity. Some eradication campaigns were conducted with force, and many lacked community involvement. These trends would continue to

haunt the efforts to eradicate malaria and smallpox in the 1950s and 1960s, despite the crumbling of most global empires and the introduction of more inclusive institutions during the latter half of the twentieth century.

HEALTH, DEVELOPMENT, AND THE LEGACIES OF COLONIALISM

World War I caused immense upheaval in the colonies and in Europe: as demand for raw materials skyrocketed during the war, a number of colonies experienced export booms that brought rapid (though still quite limited) industrialization and urbanization as well as a swelling of government coffers.⁸⁹ Ideologies shifted as the Allied powers used the language of self-determination to explain their motivations for war. Colonialism itself was called into question under these renewed democratic ideals, forcing colonizers to justify their presence in the colonies in new ways.⁹⁰

The cataclysm of World War II—and the sober discussions of building a robust, rights-based international community in its aftermath (see chapter 9)—would have powerful effects on the development of global public health. Some of this great debate was quite explicit. Early in the war, the Atlantic Charter ratified the U.S. agreement to support Britain with Roosevelt’s understanding that the end of the conflict would also spell the end of Britain’s global empire.⁹¹

As the political realities of the postwar, and increasingly postcolonial, world became clear, the political, military, and economic power asymmetries between the global North and South were reorganized around emerging languages of “development,” replete with a set of practices deeply rooted in colonial disparities. Development redefined the relationship between rich and poor countries, the colonizers and their former colonies. Although the war itself raised the key issues—and in part catalyzed the wave of independence movements across Africa, Asia, and Latin America—the language of technocracy and science was more explicitly harnessed than that of equity. Arturo Escobar, Wolfgang Sachs, and others from the academic field of development studies have traced the origins of modern development ideology to President Harry Truman’s inaugural address in 1949, during which Truman introduced his vision of a “fair deal” for the postwar world:⁹²

We must embark on a bold new program for making the benefits of our scientific advances and industrial progress available for the improvement and growth of underdeveloped areas. . . . The old imperialism—exploitation for

foreign profit—has no place in our plans. What we envisage is a program of development based on the concept of democratic fair dealing. . . . Greater production is the key to prosperity and peace. And the key to greater production is a wider and more vigorous application of modern scientific and technical knowledge.⁹³

The commitment to the welfare of the former colonies was reflected in the scale of the projects, which were often financed through official state policies, including export taxes and state control of exporting industries. The projects under this development framework were, however, heavily shaped by the legacies of colonial medicine. Given limited resources, administrators of development projects were frequently forced to choose between immediate provision of services and long-term investment in infrastructure. “Socialization for scarcity”—the assumption that resources for poverty reduction and international health initiatives will be in perpetually short supply—would come to be the dominant logic of international health, whether acknowledged or not. For example, tension arose between officials in the colonies and those in the British legislature, with the former pushing for spending on social services such as health and educational projects, and the latter asserting the importance of roads and communications (projects that would not require continuing expenditures in future years).⁹⁴ Colonial state institutions rarely extended their reach beyond a few urban centers, often on the coast.⁹⁵ As a result, development usually lagged far behind expectations and need. Moreover, when colonies gained their independence, these nascent nations were often left with incomplete infrastructures, very small national budgets based almost entirely on export crops, and enormous fiscal responsibilities to their populations.⁹⁶

Flush with the optimism of independence in the 1950s and 1960s, some new states embarked on ambitious development projects similar to those run by the colonial forces that preceded them. Because state tax revenues were generally determined by money brought in from one or two export crops or commodities—for example, cocoa in Ghana, oil in Nigeria, and tea and coffee in Kenya—the success of these projects depended on global commodity prices. Postcolonial states thus achieved varying levels of “development” at different times, but few national health systems were consistently successful and comprehensive. Throughout the 1970s and 1980s, health systems, education, and infrastructure development suffered greatly as a result of conflicts, natural disasters, declines in export prices and access to credit, sharp

political pressures, and heavy-handed postcolonial policies (discussed in the next chapter).

In addition to introducing new development frameworks globally, the end of World War II ushered in a different era of international cooperation. The new United Nations and its affiliated agencies strove to succeed where the League of Nations had failed. One of the first declarations of the United Nations, proposed by Brazil and China, requested the formation of a single agency responsible for international health cooperation. PAHO, the OIHP, the League of Nations Health Committee, and others were given the task of drafting the constitution for a unified global health body.⁹⁷ By 1948, the World Health Organization (WHO) had been formed and the first World Health Assembly had been convened, bringing together representatives from nearly seventy nations.

At its formation, two primary aims set the WHO apart from its predecessors: universal membership and decentralization.⁹⁸ The organization actively sought to engage states even before they became members of the United Nations: as of 2010, 193 countries were included in the member roster. Due in large part to the preexistence of PAHO and the OIHP, its European counterpart, and to the pressure imposed by the United States to maintain the strength of PAHO, the WHO chose a decentralized structure. It created regional offices that would carry out most of the responsibilities of the organization and a central office in Geneva to serve as a coordinating body. PAHO became the regional office for the Americas, while the OIHP was subsumed into WHO headquarters in Geneva. Integrating PAHO served to engage the United States in the WHO, which was especially important given its new role as a global superpower.

When cholera broke out in Egypt in 1947, the WHO took on its first international health crisis. The organization stepped in with both diplomatic efforts—WHO epidemiologists discouraged neighboring countries from enacting unnecessary quarantine regulations—and technical help, providing supplies for rehydration treatment, sanitation, and extensive vaccination.⁹⁹ The campaign was a remarkable success, and the WHO gained international legitimacy as an advisory and coordinating body. Nevertheless, its role as the principal funder and implementer of international health programs was not well defined until 1955, when the organization announced the most ambitious goal in its history: the global eradication of malaria.

The Malaria Eradication Programme

The parasite that causes malaria has a complex life cycle with two points of vulnerability: as a breeding parasite in the human host and as a latent source of infection carried by *Anopheles* mosquitoes. Although most effective malaria campaigns have included attempts to exploit both vulnerabilities, debate continues regarding the best way to protect humans from the illness. Again, socialization for scarcity has pitched this as an either-or debate. Those who advocate vector control encourage drainage of potential mosquito breeding sites, the use of insecticides in homes and fields, and widespread distribution of insecticide-treated bed nets. Many argue that vector control can take place even in areas without robust health services and that the full cooperation of at-risk populations is not required, a feature of the approach that was also attractive to colonial officials in the early twentieth century.

Parasite control, in contrast, relies on medicinal intervention in the form of malaria prophylaxis and treatment with such medications as quinine and, more recently, artemisinin. Proponents of this strategy cite the immense and prohibitive cost of draining all mosquito breeding sites. A third perspective, typically less popular within the medical community, argues that malaria will never be effectively controlled without transformation of living conditions and land tenancy agreements that keep peasant farmers poor—and thus less likely to invest in land improvements such as drainage.¹⁰⁰

In the postwar period, such debates took place in a rapidly changing social and political environment, as a properly biosocial analysis reveals. But changes were also occurring in the pathogens, vectors, and tools to fight them. The years following World War II saw not only the end of colonization and an increase in international cooperation but also the development of new antibiotics and the pesticide DDT. These powerful compounds transformed the public's opinion of science in general and medicine in particular: that technological tools could be so effective against great scourges gave the world a strong belief in the power of science and technology.¹⁰¹ Skepticism over the role of these technical fixes would arise later, but general faith in technology boomed in the immediate postwar years.

The WHO's Malaria Eradication Programme (MEP) can be viewed as a direct result of the success of DDT and vector control as well as a result of donors' preferences for narrow, top-down approaches to

health. DDT had been used during World War II to protect troops from mosquitoes and afterward to control disease among displaced populations in Europe. Reflecting a global faith in the power of DDT, which silenced much potential debate about vector versus parasite control, the WHO announced in 1955 that it would coordinate and fund a program designed to completely eradicate malaria around the world.¹⁰² Success in several small-scale campaigns, including the Panama Canal sanitation efforts, had strengthened belief in vector control. The Cold

The McKeown Hypothesis

Thomas McKeown, a physician and influential demographic historian, has challenged the extent to which medical interventions have contributed to improvements in health outcomes. In his research, published from the 1950s to the 1980s, McKeown classifies the variables contributing to the control of infectious disease into three categories: “medical measures (specific therapies and immunization), reduced exposure to infection, and improved nutrition.” He argues that declining mortality can be attributed mainly to improved standards of living, which lead to reduced exposure to disease and, more important, advancement in nutrition. Contrary to the widely held belief that advances in medicine have been a main cause of population growth, McKeown suggests that medical measures played only a part, perhaps a small one, in improving health outcomes (see figures 3.9 and 3.10, which show steep declines in mortality rates before the introduction of medical measures to combat several diseases).¹

McKeown’s thesis has since been criticized for its methodology and for the political biases that may have influenced his argument.² Despite the technical flaws of McKeown’s study, however, his ideas continue to resonate. McKeown forces us to think about the limitations of relying too heavily on targeted medical interventions, and he asks us, albeit indirectly, to examine broader approaches that consider the social, political, and economic factors of the local context. From this perspective, one can better understand the failure of the malaria eradication campaign and others that have ignored crucial dimensions that pattern disease persistence.

McKeown’s work has remained relevant, but it was challenged significantly by the course of the response to the AIDS epidemic (see figure 3.11). In the span of three decades, scientists developed preventative tools, diagnostics, and powerful therapeutics that directly led to substantial and observable reductions in AIDS mortality in a wide variety of contexts.

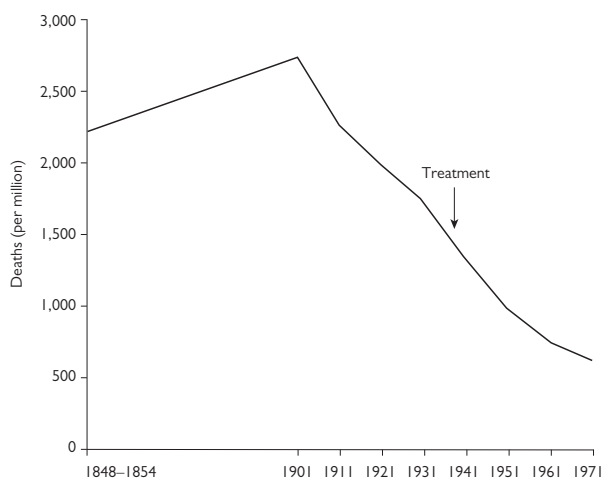


FIGURE 3.9. For bronchitis, pneumonia, and influenza in England and Wales from 1848 to 1971, McKeown and colleagues present data showing that dramatic declines in the mortality rate preceded major biomedical interventions. (These death rates are standardized to the age-sex distribution of the 1901 population.) Source: Thomas McKeown, R. G. Record, and R. D. Turner, "An Interpretation of the Decline of Mortality in England and Wales during the Twentieth Century," *Population Studies* 29, no. 3 (1975): 391-422.

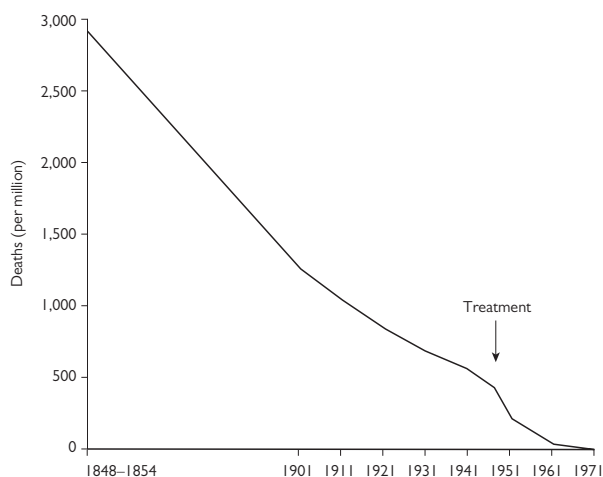


FIGURE 3.10. For cases of respiratory tuberculosis in England and Wales from 1848 to 1971, McKeown and colleagues present data showing a precipitous decline in the mortality rate well before major biomedical interventions were available. (These death rates are standardized to the age-sex distribution of the 1901 population.) Source: Thomas McKeown, R. G. Record, and R. D. Turner, "An Interpretation of the Decline of Mortality in England and Wales during the Twentieth Century," *Population Studies* 29, no. 3 (1975): 391-422.

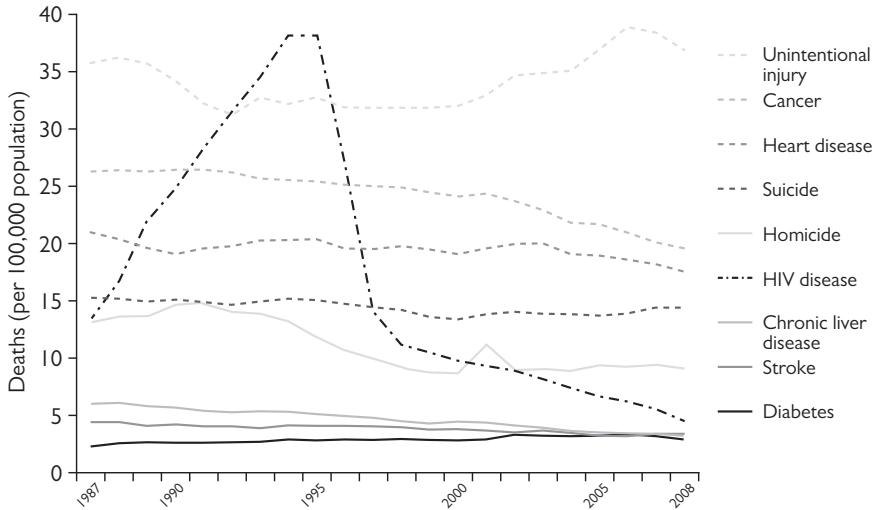


FIGURE 3.11. The case of HIV can be taken as a direct challenge to the McKeown hypothesis: AIDS mortality rates in the United States declined dramatically after the development of Highly Active Antiretroviral Therapy (HAART) in 1995. Source: Centers for Disease Control and Prevention, *HIV Mortality: Trends (1987–2008)*, “Slide 18: Trends in Annual Rates of Death Due to the 9 Leading Causes among Persons 25–44 Years Old, United States, 1987–2008,” www.cdc.gov/hiv/topics/surveillance/resources/slides/mortality/slides/mortality.pdf. Courtesy Centers for Disease Control and Prevention.

War may have also played a role in favoring an eradication campaign over a more comprehensive rural development plan that would have facilitated malaria control: wary of the ideologies underpinning a program seeking to organize and serve rural masses, political leaders in the United States and other Western powers tended to prefer a more vertical, disease-specific approach.¹⁰³

Although the Malaria Eradication Programme included some distribution of chloroquine, an antimalarial, the MEP favored the tactic of spraying DDT inside homes. Every home in malaria-stricken regions was to be sprayed at least once a year until the disease was eradicated. Even considering DDT’s effectiveness and ease of application, this program was slated to be an enormous undertaking. Spraying teams around the world faced tremendous obstacles, including accessing very remote areas, earning the cooperation of residents, anticipating the possible reintroduction of the parasite from neighboring regions that lacked robust malaria eradication programs, and paying the sky-

rocketing costs of personnel and supplies. The program was essentially top-down in nature. Local teams received orders from the WHO and did not have the flexibility to adapt them to local social or geographic conditions or to address problems that arose during the campaign, such as distrust between local populations and spray teams, or labor migration patterns that periodically reintroduced the parasite into previously malaria-free areas.¹⁰⁴ There were other forms of resistance, too. By the mid-1960s, mosquitoes and the malaria parasite were demonstrating resistance (to DDT and chloroquine, respectively), and the future of malaria eradication began to look extremely expensive, if not outright impossible.

When the WHO abandoned the Malaria Eradication Programme in 1969, opting for a less ambitious program of malaria control, eighteen countries that had participated declared eradication, and eight more followed in the next few years. As historian Randall Packard points out, however, every one of these countries was either developed (such as the United States and Spain), an island (including much of the Caribbean), or socialist (primarily Eastern European nations).¹⁰⁵ He argues that the primary reason for the failure of the MEP was not lack of resources or mosquito resistance to DDT, nor was it the calls from environmentalists to end the use of DDT, which were not heeded until well after the WHO had abandoned the MEP. Rather, the campaign failed, writes Packard, because of overwhelming belief in technological fixes. The WHO eradication campaign underestimated the biosocial fact that malaria is a disease deeply embedded in social factors like agricultural traditions (including the use of irrigation reservoirs) and labor migration patterns; a campaign like the MEP could succeed only when effective diagnostic, preventative, and therapeutic tools were linked to a transformation of living conditions in the poor countries in which malaria remains, to this day, a leading killer of children and adults.

Smallpox Eradication

Not all eradication campaigns were doomed to fail. The WHO initiated the smallpox eradication campaign in 1967 as its efforts against malaria were winding down. The smallpox program focused on two main activities: large-scale vaccination, and surveillance and containment. The vaccination campaigns aimed to reach 80 percent of the

population of endemic areas, both by convening large gatherings of local residents at a single vaccination point and by sending vaccination teams from house to house. For surveillance and containment, a strong reporting infrastructure was necessary to enable local health workers to notify a country's smallpox-control program when ill residents were identified.¹⁰⁶ Once a case was discovered, health workers were expected to isolate the sick, quarantine those who had come into contact with infected people, ensure vaccination of the whole community, and disinfect all surfaces that might have been contaminated with the virus.¹⁰⁷ This system required large numbers of health workers, but not necessarily professionals with advanced training. In fact, the vast majority of people engaged in this campaign were trained quickly and paid very little.¹⁰⁸

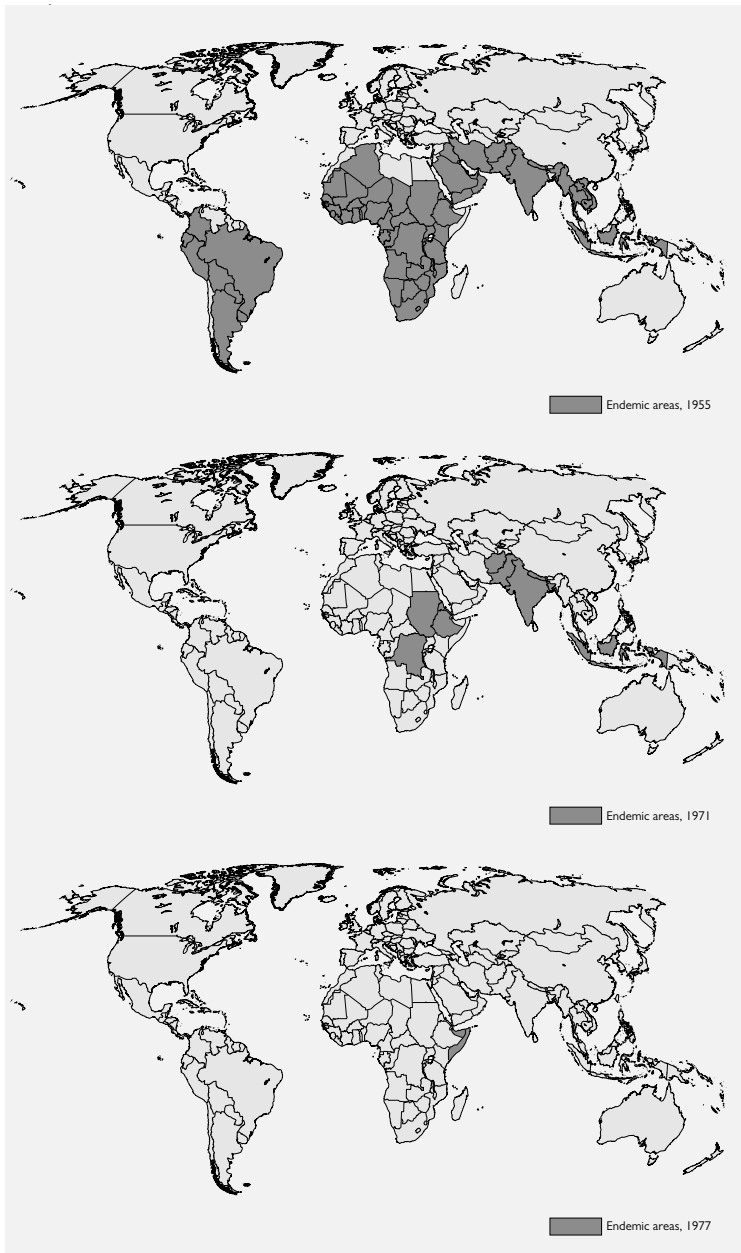
The eradication of smallpox is lauded as a triumph of modern public health efforts, in part because of superior program management and design. There were significant obstacles, such as detecting smallpox sufferers in rural areas, where cases often went unreported. Patients would frequently not seek care even after symptoms developed, primarily because no known treatment for the disease existed (and of course because modern biomedical care, with little to offer, was only scantily available in smallpox-affected regions). As with malaria, migration created the risk of reimporting the disease into previously eradicated areas. Distributing vaccines and training health workers across broad swaths of the developing world also posed major logistical challenges for implementers to overcome. In addition to remarkable program design and management, several external factors enabled success as well: smallpox is easily identifiable, symptoms appear quickly, transmission is easily tracked, and the vaccine is effective and easy to administer (see table 3.1).¹⁰⁹ In 1977, it was announced that smallpox had been eradicated globally (see map 3.3).

Despite its success, the campaign was not without controversy. In three of the last countries to declare eradication—India, Bangladesh, and Ethiopia—health workers met strong resistance from local populations. In fact, multiple incidents of coercion, sometimes violent and often under the auspices of expatriate health officials commissioned by the WHO to coordinate campaigns, were reported during the end of the campaign in South Asia. Besides being a human rights concern, forced vaccinations left a residue of resentment in rural communities, potentially hampering future public health efforts.¹¹⁰

TABLE 3.1 FACTORS AFFECTING THE SUCCESS OF ERADICATION EFFORTS: MALARIA VERSUS SMALLPOX

	Malaria	Smallpox
Transmission	Vector-borne: <i>Anopheles</i> mosquitoes	Direct personal contact
Recurrence	Patients may experience recurrent bouts throughout their lifetime	No recurrence; immunity is acquired after a single infection or vaccination
Latency	Can remain in the body without symptoms for several months	Symptoms appear within ten to fourteen days
Case-Finding	Diagnosis can be challenging; presentation mirrors other diseases	Infection is marked by obvious and distinctive sores on the body
Prevention	Either vector control (bed nets, DDT spraying) or parasite control (chemoprophylaxis)	Vaccine is extremely effective

Nevertheless, the success of the WHO's campaign against smallpox restored the agency's political stature, which had been damaged by the failure of malaria eradication. Historical reflection suggests that the smallpox campaign may, in the eyes of some, be considered insufficient, since in many regions it failed to build infrastructure for long-term impact, as demonstrated in subsequent vaccination drives and other public health efforts. The top-down approach of the campaign, which prioritized technological fixes over consideration of local context and broader infrastructure, and which at times compromised the agency of the populations targeted for intervention, would later be critiqued by advocates of community-based primary health care. To some extent, the smallpox campaign allowed a continuation of the discourses of colonial medicine, such as underlying notions of the diseased native, a pathologized collective, and faith in technological fixes. All of that said, however, the world had been freed of a terrible scourge. Program implementers achieved a goal that many critics believed would be unattainable and overcame numerous logistical and political challenges at each stage. Smallpox had been responsible for 300 million to 500 million deaths in the twentieth century, and the last naturally occurring outbreak occurred on October 27, 1977.¹¹¹ For many, the eradication of smallpox remains the high-water mark of twentieth-century global health efforts.



MAP 3.3. The progress of the smallpox eradication campaign, 1955–1977. In 1955, much of the global South faced a heavy smallpox burden. The WHO’s eradication campaign began in 1967, and by 1971 the disease had been eliminated from many countries. Somalia was the last country to eliminate smallpox before global eradication was declared in 1977. Courtesy Centers for Disease Control and Prevention Image Library.

CONCLUSION

To discern the persistence of colonial health structures in modern-day global health practices is to understand the importance of historical analysis in tracing both the continuities and ruptures between present and past public health practices. Throughout this book, we argue that the resocializing disciplines can illuminate—and improve—global health practice. History provides a vital, easily available, yet frequently overlooked method of critical social analysis, which can shed important light on the political, economic, cultural, intellectual, and material specificity of the global health problems we now face. Historical consciousness of the colonial roots of global health challenges us to question the knowledge frameworks that constitute the emerging field of global health today.

If there is one message to retain from this chapter, it is that history offers many lessons for modern global health initiatives. Critical social history seeks to accomplish three goals: to render the present unfamiliar, and therefore open to social critique; to emphasize the role of *continuity* between, for example, current global health efforts and former colonial rule; and to work to understand the rifts and discontinuities that account for meaningful social change.

The knowledge frameworks carried forward from colonial times continue to influence both who is invited to the policymaking table and how global health agendas are then prioritized. As the next chapter argues, such frameworks often put a premium on vertical programs, such as eradication campaigns, at the expense of long-term efforts to strengthen health systems; they also may have little regard for the social and economic roots of disease or for interventions designed to address them.

Chapter 4 details a series of crises that arose in the 1970s. In light of the changing international politics of the postcolonial era and the failure of large-scale interventions like the Malaria Eradication Programme, the dominant models of international health and international development were called into question. The reformulation of global public health priorities in the 1970s led to broader international discussions of “horizontal” health interventions and the birth of the primary health care movement.¹¹²

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